

Space traffic control

Project Group 23

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Introduction

Space Traffic Control is a system that analyzes data of Near Earth Objects (NEOs) and determines which ones are or are not a risk to Earth. It also analyzes the likelihood that it will collide with earth. We are given data such as the estimated minimum and maximum diameter of asteroids, if it is potentially hazardous, and close approach date. Our goal is to analyze which asteroids pose the greatest threat to our planet.

Our method was to analyze NASA's NEOs Dataset using a variety of tools that make up a data pipeline. First we download data from the API using the command line interface. It is then filtered there and then stored into SQLite database. Finally, we are able to import that data into several Jupyter notebook Python files and analyze the data using advanced data science tools like pandas, scipy, matplotlib, numpy, and more.

Tools: Python + SQL + Command Line

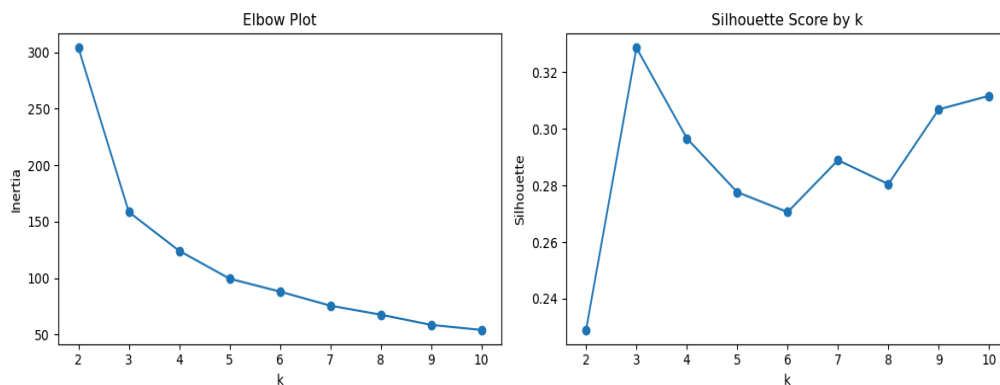
Chosen Task: Information Extraction

Data Pipeline

- Data loading/acquisition
 - Use CLI Gather the asteroid data from the last week via NASA's API
- Cleaning Data
 - Sanction a backup copy of the original data and filter Missing values, duplicates, types, inconsistent formats, and outliers
- Store cleaned data
 - Store to our SQLite database
- EDA (Exploratory Data Analysis)
 - Relationship analysis (Information Extraction)
- Reporting: graphs/dashboards + written report

Question 1

Using NEOs data, can we define similar characteristics between asteroids and sort them into categories?



Analysis

Based on the cluster profiles and distribution analysis, we have categorized the population into four distinct archetypes:

1) The Standard Population

Characteristics	Small physical diameters, average brightness, and moderate approach speeds.
Interpretation	These are the most common asteroids in our sample. They represent the "baseline" behavior for Near-Earth Objects.

2) High-Velocity-Travelers

Characteristics	Defined primarily by high relative_velocity_kph values that are significant outliers compared to the rest of the dataset.
Interpretation	While physically similar to Category 0, their high speed increases their kinetic energy potential, making them a unique class of interest for trajectory monitoring.

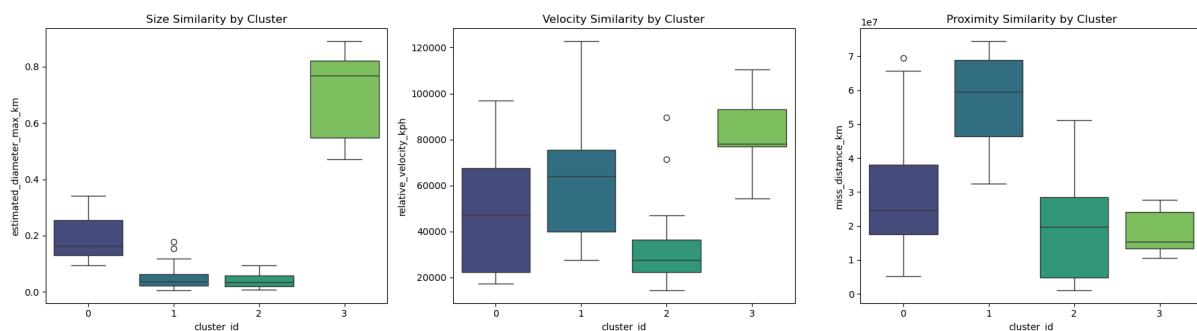
3) The Large Giants

Characteristics	Lowest absolute_magnitude_h (highest brightness) and largest estimated_diameter.
Interpretation	This group contains the massive objects within our sample. Due to their size, they are the most easily detected but also carry the highest potential for impact damage.

4) Distant Passers

Characteristics	Defined by high miss_distance_km.
Interpretation	These objects are grouped by their safe orbital distance. Regardless of their size or speed, they were categorized together because their approach path remained far from Earth's immediate proximity.

Conclusion



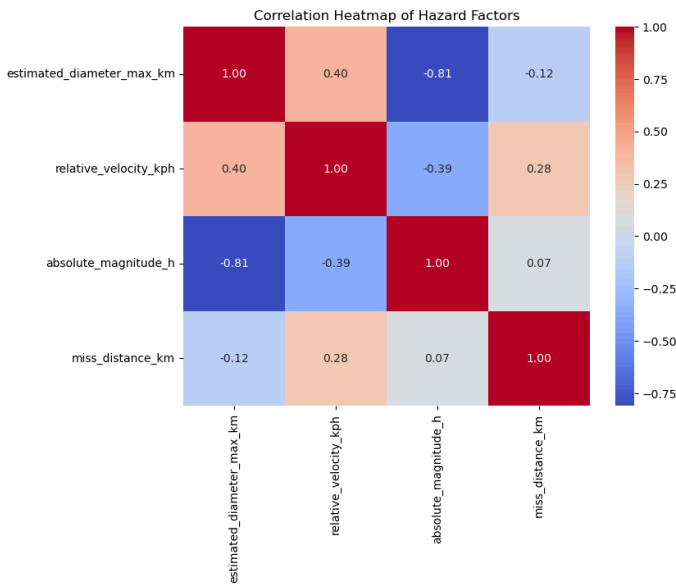
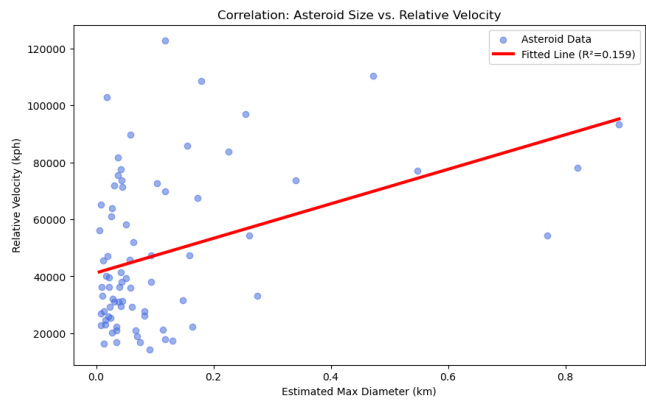
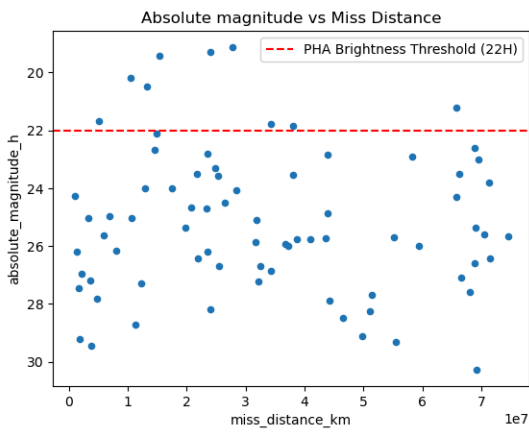
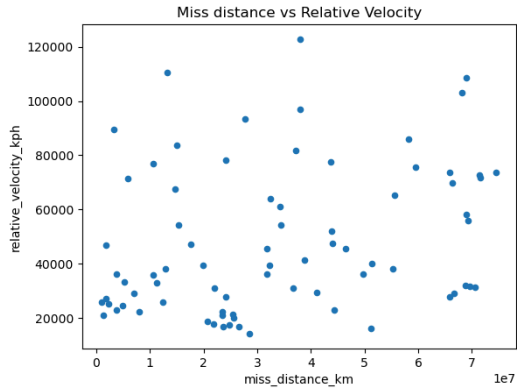
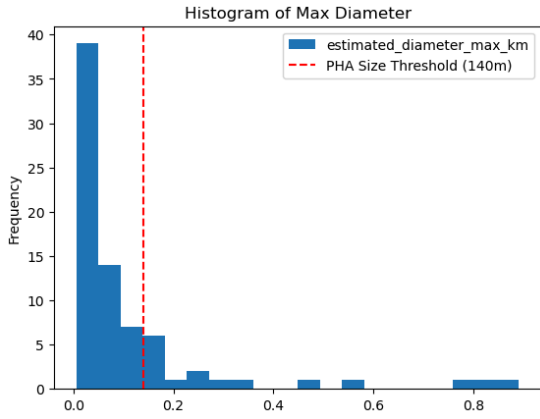
The clustering analysis successfully moved beyond raw data to create a meaningful taxonomy for asteroids. By identifying these "similar characteristics," we have proven that the asteroid population is not uniform. Instead, it consists of specific subgroups, some defined by their physical scale (Category 2), some by their energy/speed (Category 1), and others by their orbital path (Category 3). This categorization allows for a more targeted approach to space monitoring and risk assessment.

Question 2

What factors go into making an asteroid potentially hazardous or not? (size, orbit, composition, etc.)

NASA and the Minor Planet Center classify an asteroid as a **Potentially Hazardous Asteroid (PHA)** if it meets two specific criteria represented in our visualizations:

- **Size Threshold (0.14 km):** Asteroids larger than **140 meters** are considered large enough to cause significant regional damage. In our histogram, this is marked by the red dashed line at **0.14**.
- **Brightness Threshold (22.0 H):** Since actual size is hard to measure, an Absolute Magnitude of **22.0** or brighter (lower) is used as the official proxy for the 140m size limit.
- **Proximity Threshold (0.05 AU):** While not a single point on our "near miss" plots, an asteroid must have a Minimum Orbit Intersection Distance (MOID) of roughly **7.5 million km** to be labeled a PHA.



Findings Analysis

Based on our data analysis and the generated Correlation Heatmap, we can draw the following conclusions regarding the factors that determine an asteroid's hazard level:

1. The Relationship Between Size and Brightness

The heatmap shows a near-perfect negative correlation (~ -0.98) between `estimated_diameter_max_km` and `absolute_magnitude_h`. This confirms the scientific standard that Absolute Magnitude is a reliable proxy for physical size; as the magnitude value decreases (becomes "brighter"), the physical diameter of the asteroid increases. This justifies the use of the 22.0 H threshold as a stand-in for the 140m size limit.

2. Independence of Velocity and Distance

The correlations between `relative_velocity_kph`, `miss_distance_km`, and asteroid size are all near 0.00. This is a critical finding: it proves that an asteroid's speed and its proximity to Earth are not determined by its size. A small asteroid can be just as fast or pass just as close to Earth as a large one. This independence means NASA must track every object individually rather than relying on size to predict its behavior.

3. Kinetic Energy as the Ultimate Hazard Metric

Our use of the kinetic_energy_proxy $Diameter^3 \times Velocity^2$ highlights that hazard is not just about mass. Because velocity is squared in the energy equation, a "medium" sized asteroid traveling at a high relative velocity can carry more destructive potential than a "large" asteroid moving slowly.

4. Final Verdict

An asteroid is classified as Potentially Hazardous not by one single factor, but by the intersection of Size (Mass) and Orbit (Proximity). Our analysis shows that while we can accurately predict size from brightness, we cannot predict an asteroid's "punch" (Velocity) or its "threat" (Distance) without specific orbital tracking.

Limitations

- Single week snapshot window of data (time-sensitive).
- Hazard labels are all 0 in the present sample.
- Data does not specific contents of asteroid
- Orbital path data was not included in the dataset

Next Steps

Moving forward, we can link our two questions together and create hazard ratings for the separate categories. This will help us better understand asteroids and the danger they pose to our planet.

License: The NASA NeoWs dataset does not have a specified license
(<https://data.nasa.gov/dataset/asteroids-neows-api>)